Mapping Nighttime Skyglow and Streetlight Glare Across Localities during Festivals: Effects on Star Visibility, Dry Eyes, Children, and Older Adults

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Abstract

Outdoor artificial light at night has grown into a widespread environmental and public health problem that obscures the night sky, wastes municipal electricity, and causes visual discomfort for sensitive groups. This study presents a spatial photometric survey conducted across six urban and residential locations in Bandel, Chinsurah, and Chandannagar in the Hooghly district of West Bengal during the peak of the Durga Puja festival. Using calibrated smartphone light sensors, Sky Quality Meter conversions, naked-eye star counts, and Bortle Dark-Sky Scale classifications, this investigation mapped local skyglow gradients and audited outdoor streetlights for shielding geometry and color temperature. High-density commercial centers such as Bandel More and Vineet Shopping City reached Bortle Class 8 to 9 (0.290 to 0.315 Lux; 17.90 to 18.12 mag/arcsec²), where 85% of audited fixtures were unshielded cool-white floodlights rated at 5000 K or higher. In contrast, a shaded residential neighborhood at Sahebbagan Boy's Club located 603 meters from Bandel More maintained Bortle Class 3 skies (0.012 Lux; 20.95 mag/arcsec²) because 90% of its lamps were warm-white fixtures rated at or below 3000 K with downward shielding[1]. Beyond atmospheric Rayleigh scattering, this paper examines how unshielded blue-rich lighting scatters across unstable tear films in people with dry eyes to trigger trigeminal nerve pain, passes through the clear lenses of young children to suppress up to 88.2% of evening melatonin, and creates blinding glare for older adults with cataracts[3]. Practical solutions are presented, including personal tear-film protection, low-cost do-it-yourself aluminium lamp shields and amber filters costing under 150 INR, and three budget-neutral municipal lighting rules[1].

*KEYWORDS:*Light pollution;Skyglow;Naked-Eye Limiting Magnitude(NELM);Rayleigh scattering;Atmospheric optics, Night sky photometry

1. Introduction

1.1 Nighttime Light Pollution and Festive Illumination in Hooghly

Over the past several decades, the rapid switch to bright solid-state outdoor lighting has changed the night environment across the world (Falchi et al., 2016; Kyba et al., 2017)[1]. Satellite measurements by Kyba et al. (2017) show that the artificially lit surface of the Earth has grown in both brightness and geographic area by more than 2% per year, leaving over 80% of the global population under light-polluted skies (Falchi et al., 2016)[1]. While street lighting is necessary for road safety, evening commerce, and pedestrian security, poorly aimed fixtures send a large share of their light straight up into the sky or horizontally into nearby homes[1]. This wasted light creates urban skyglow—a bright dome of scattered light that hides the stars, disturbs nocturnal birds and insects, and wastes municipal electricity[1].

In the riverfront towns of West Bengal—specifically Bandel, Chinsurah, and Chandannagar in the Hooghly district—everyday street expansion meets intense seasonal lighting during the autumn Durga Puja and Jagaddhatri Puja festivals[1]. Chandannagar is famous across India for its decorative street-lighting artistry, while Bandel and Chinsurah serve as busy railway and commercial centers[1]. During late October, thousands of temporary bamboo-and-cloth pandals, historic churches, shopping streets, and road junctions are lit up with bare halogen lamps, high-wattage light-emitting diode (LED) floodlights, and bright commercial signs[1]. Rather than treating light pollution as an unavoidable side effect

of town growth, this study treats urban skyglow as a solvable engineering and public health problem caused by two controllable factors: unshielded fixtures that send light above the horizontal plane, and blue-heavy white bulbs with high Correlated Color Temperatures (5000 K to 6500 K)[1].

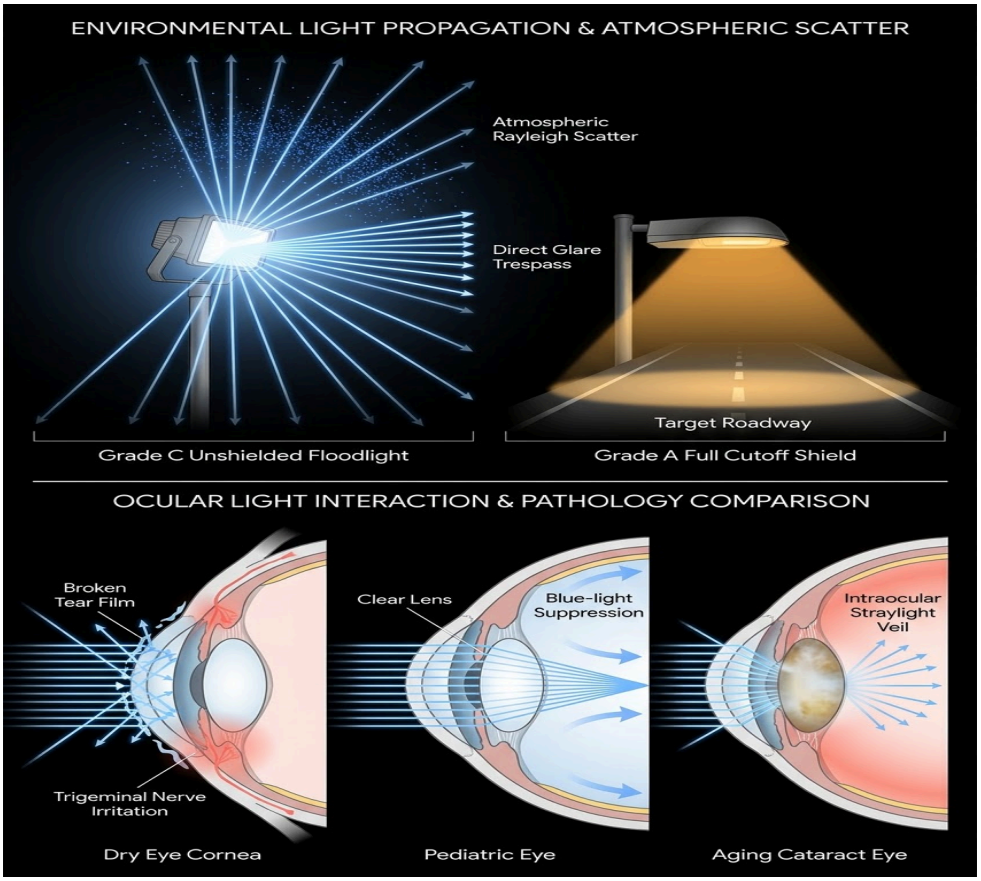
1.2 Why Bright Nighttime Lights Hurt People with Dry Eyes, Young Children, and Older Adults

Most municipal lighting surveys focus only on astronomy or electricity bills and ignore how bare, bright lights affect human eyes at night[1]. In dim evening conditions, the human pupil naturally opens wider (dilates) to let in more light[10]. When a dilated eye looks toward an unshielded floodlight or bare streetlamp, light rays enter through the outer edges of the cornea and lens, exposing small optical imperfections that do not bother a person during the daytime when the pupil is small[10]. Walking through brightly lit streets at night creates serious visual strain and physical discomfort for three groups in our community: people with Dry Eye Disease (including the author of this study), young children, and older adults[3].

For anyone living with Dry Eye Disease, stepping into a brightly lit commercial junction at night causes immediate visual blurring and eye pain[3]. In a healthy eye, a thin, smooth layer of tears made of oil, water, and mucus covers the cornea, focusing incoming light cleanly while protecting surface nerve endings[3]. When a person has dry eyes—either because their tear glands do not produce enough moisture or because dry autumn air, wind, and long hours of reading or computer work cause tears to evaporate too fast—the tear film breaks apart into uneven dry patches across the cornea[3]. Instead of focusing to a sharp point on the retina, light rays from bare streetlamps and floodlights hit the rough, dry corneal surface and scatter in all directions[9]. This forward light scatter creates glowing rings (halos) and spiky starburst patterns around every streetlight and car headlight, washing out contrast and making it hard to see steps or uneven pavement[9]. At the same time, the broken tear film leaves corneal nerve endings connected to the trigeminal nerve exposed to the air[3]. When harsh blue-white light strikes these exposed nerves, it triggers sharp light sensitivity (photophobia), stinging, burning, and reflex tearing[3].

Young children face a different biological risk when exposed to bright artificial light at night[4]. A young child's eye has a wider pupil and a very clear crystalline lens that has not yet developed the natural yellow pigment that protects adult eyes[6]. Because of this clear lens, a much larger fraction of short-wavelength blue light (440 nm to 480 nm) reaches a child's retina when streetlight glare shines into a bedroom window or illuminates an evening festival walk[6]. Once this blue light hits specialized light-sensitive cells in the retina, it signals the brain to stop producing melatonin, the hormone needed for sleep, while raising alertness and cortisol levels[4]. Clinical studies show that under identical evening lighting (580 Lux), melatonin levels drop by 88.2% in children—nearly double the suppression seen in adults[6]. This explains why bright, cool-white street and festival lights make young children restless and disrupt their sleep schedules[4].

Older adults also struggle in brightly lit night environments because of natural aging in the eye's lens[5]. Over time, proteins in the crystalline lens clump together to form cataracts, or the cornea swells from conditions such as Fuchs' dystrophy[8]. Instead of letting light pass straight through to the retina, a cloudy lens scatters incoming light inside the eyeball—a clinical problem known as intraocular straylight[8]. When an older person walks past an unshielded floodlight or faces oncoming traffic at night, this scattered straylight spreads a bright haze across the retina, hiding curb edges, road markings, and pedestrians in the shadows[5]. It also takes older eyes much longer to adjust when moving from a blindingly bright intersection back into a darker lane[5]. Figure 1 illustrates how unshielded blue-rich lighting drives both upward atmospheric skyglow and internal light scattering across dry-eye tear films, pediatric lenses, and older cataractous eyes[1].



2. Methods

2.1 Physical Theory and Mathematical Equations

To measure night sky brightness with the same mathematical precision used in space-telescope stellar photometry (Borucki et al., 2010; O'Donovan et al., 2006; Seager & Mallén-Ornelas, 2003)[1], ground-level light surveys must connect two measurement systems: physical photometry (which measures ground illuminance in Lux and sky luminance in candelas per square meter, cd/m²) and astronomical brightness scales (which measure sky darkness in magnitudes per square arcsecond, mag/arcsec², and star visibility using Naked-Eye Limiting Magnitude, NELM)[1].

In astronomy, apparent brightness follows Pogson's Ratio, a logarithmic scale where a difference of 5 magnitudes equals a factor of 100 in luminous flux (F)[1]:

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right)$$

Here, m_1 and m_2 are the apparent magnitudes of two light sources or patches of night sky, and F_1 and F_2 are their physical brightness fluxes[1]. Because the formula uses a negative sign, larger magnitude numbers indicate fainter light and darker skies[1].

When measuring zenith sky darkness (μ_{sky}, in mag/arcsec²) with a Sky Quality Meter (SQM) or a calibrated smartphone camera sensor, we convert surface brightness into physical luminance (L, in cd/m²) using the standard radiometric equation[1]:

$$L = 10.8 \times 10^4 \times 10^{-0.4\mu_{\text{sky}}}$$

When a light meter records luminance L, the inverse equation solves for zenith Sky Quality (μ_{sky})[1]:

$$\mu_{\text{sky}} = -2.5 \log_{10}\left(\frac{L}{10.8 \times 10^4}\right) = 12.583 - 2.5 \log_{10}(L)$$

To connect sky brightness readings (μ_{sky}) with the faintest star a human observer can see at the zenith (NELM), we apply the human contrast-threshold equation derived by Crumey (2014)[1]:

$$\text{NELM} \approx 7.93 - 5 \log_{10}\left(10^{4.316 - 0.2\mu_{\text{sky}}} + 1\right)$$

As artificial skyglow brightens the background sky, μ_{sky} drops, washing out faint stars and lowering the NELM count (Crumey, 2014)[1]. For observers with dry eye syndrome or cataracts, light scattering inside the eye reduces contrast even further, lowering their personal star-visibility limit below Crumey's baseline[1].

The main physical process that turns upward streetlight beams into a wide skyglow dome is Rayleigh scattering by nitrogen (N_2) and oxygen (O_2) molecules in the air[1]. The Rayleigh scattering cross-section $\sigma_s(\lambda)$ per molecule is inversely proportional to the fourth power of the light's wavelength (λ^{-4})[1]:

$$\sigma_s(\lambda) = \frac{8\pi^3 (n^2 - 1)^2}{3N\lambda^4}$$

In this equation, n is the refractive index of air, N is the number density of air molecules, and λ is the wavelength of light[1]. Many commercial floodlights in Bandel and Chandannagar use cheap, cool-white LEDs (5000 K and above) that emit a strong blue spike around $\lambda_{\text{blue}} \approx 450$ nm, whereas warm-amber Dark-Sky compliant LEDs (3000 K and below) peak around $\lambda_{\text{amber}} \approx 600$ nm[1]. Comparing their scattering rates gives[1]:

$$\frac{\sigma_s(450\text{ nm})}{\sigma_s(600\text{ nm})} = \left(\frac{600}{450}\right)^4 = \left(\frac{4}{3}\right)^4 = \frac{256}{81} \approx 3.16$$

Short-wavelength 450 nm blue light scatters 3.16 times (316%) more strongly in the air than 600 nm warm-amber light[1]. This same fourth-power scattering law explains why 450 nm blue light scatters so severely across the rough tear film of a dry eye and inside a cloudy cataract lens[1].

Finally, for a streetlight of intensity $I(\theta)$ mounted at height h above a street, the useful horizontal illuminance E_h reaching the road at angle θ from the vertical is given by the cosine-cubed law[1]:

$$E_h = \frac{I(\theta)\cos^3\theta}{h^2}$$

When light is emitted near the horizontal plane ($\theta \to 90^\circ$), $\cos^3\theta$ approaches zero, meaning those rays provide almost no useful light on the ground and instead shine straight into pedestrians' eyes and upstairs windows[1]. Any light emitted at or above the horizontal plane ($\theta \geq 90^\circ$) goes straight into the sky[1]. We measure this wasted fraction using the Upward Light Output Ratio (ULOR)[1]:

$$\text{ULOR} = \frac{\Phi_{\text{up}}(\theta \geq 90^\circ)}{\Phi_{\text{total}}} \times 100\%$$

Here, $\Phi_{\text{up}}(\theta \geq 90^\circ)$ is the light flux sent above 90° and Φ_{total} is the total light output of the fixture[1].

Table 1: Summary of Photometric and Optical Equations

Equation Name	Formulae	Output Units
Eq. 1: Pogson's Magnitude Ratio	$m_1 - m_2 = -2.5\log_{10}\left(\frac{F_1}{F_2}\right)$	mag (Dimensionless)[1]
Eq. 2: SQM to Luminance Conversion	$L = 10.8 \times 10^4 \times 10^{-0.4\mu_{sky}}$	cd/m²[1]
Eq. 3: Luminance to SQM Solver	$\mu_{sky} = 12.583 - 2.5\log_{10}(L)$	mag/arcsec²[1]
Eq. 4: Crumey's NELM Formula	$NELM = 7.93 - 5\log_{10}\left(10^{4.316-0.2\mu_{sky}} + 1\right)$	mag (NELM)[1]
Eq. 5: Rayleigh Scattering Law	$\sigma_s(\lambda) = \frac{8\pi^3(n^2-1)^2}{3N\lambda^4}$	m²/molecule[1]
Eq. 6: Blue vs. Amber Scatter Ratio	$\frac{\sigma_s(450\,nm)}{\sigma_s(600\,nm)} = \left(\frac{600}{450}\right)^4 = \frac{256}{81} \approx 3.16$	Ratio (Dimensionless)[1]
Eq. 7: Cosine-Cubed Road Illuminance	$E_h = \frac{I(\theta)\cos^3(\theta)}{h^2}$	Lux (lm/m²)[1]
Eq. 8: Upward Light Output Ratio	$\text{ULOR} = \frac{\Phi_{\text{up}}(\theta \geq 90^\circ)}{\Phi_{\text{total}}} \times 100\%$	Percentage (%)[1]

2.2 Field Survey Protocol and Study Sites

Field measurements were collected across six locations in Bandel, Chinsurah, and Chandannagar on clear, cloudless nights during the New Moon window of October 24–25, 2026, between 11:15 PM and 1:30 AM Indian Standard Time (IST)[1]. Taking readings more than 60 minutes after sunset during the New Moon removed any sunlight or moonlight, ensuring that all measured skyglow came from streetlights, commercial signs, and Durga Puja pandal lighting[1].

As outlined in Table 2, a five-step protocol was followed at each site[2]. First, the observer waited 10 to 15 minutes at each location using a dim red screen filter so their eyes could adapt to the dark[1]. Second, zenith sky brightness was measured by holding the calibrated smartphone light sensor horizontally at chest height (1.3 meters above the ground), pointing

straight up at the zenith ($\theta = 0^\circ$) away from tree branches or building roofs[1]. Five readings of horizontal illuminance (Lux) and sky surface brightness (mag/arcsec²) were taken and averaged at each site to filter out passing car headlights[1]. Third, Naked-Eye Limiting Magnitude (NELM) was recorded by finding the faintest stars visible to the naked eye in overhead constellations (Ursa Major, Pegasus, and Cassiopeia)[1]. Fourth, outdoor light fixtures within 50 meters were audited and graded by their shielding angle (Grade A, B, or C) and color temperature in Kelvin[1]. Fifth, two verification photographs—one upward photo of the night sky and one close-up photo of the nearest dominant streetlight or floodlight—were taken at every station[2].

Table 2: Five-Step Nighttime Field Survey Protocol

Step Number	Field Task	Standard Procedure Followed	Output Record
Step 1	Standardize Sky Conditions	Take readings at least 60 minutes after sunset (11:15 PM to 1:30 AM IST) on clear, cloud-free New Moon nights[1].	Field Weather and Time Log[2]
Step 2	Record Zenith Brightness and Star Counts	Measure zenith illuminance (Lux), SQM (mag/arcsec²), and faintest visible stars (NELM) at Sites 01 to 06[1].	Master Survey Data Log (Table 6)[1]
Step 3	Audit Streetlights and Take Photos	Grade nearest lamps by shielding (Grades A, B, C) and Kelvin rating; take 1 sky photo + 1 lamp photo per site[1].	Photo Verification Matrix (Table 8)[2]
Step 4	Build Spatial Map Layer	Plot GPS coordinates on Google My Maps using the color-coded Bortle pin scale (Green, Yellow, Orange, Red)[1].	Interactive GIS Map Layer[1]
Step 5	Draft Municipal Action Plan	Group site data by urban zone, test DIY lamp shields, and write a 1-page civic policy brief[1].	Municipal Dark-Sky Policy Brief[1]

2.3 Software Tools and Calibration Scales

To keep the project reproducible for high school researchers without expensive laboratory hardware, this study combined free mobile sensing apps, web-based Geographic Information System (GIS) mapping, and a custom Java data-processing program (adapted from the author's Track 1 ExoplanetAnalyzer.java pipeline)[1]. Table 3 lists each software tool used in the

study[1].

Table 3: Software Programs Used for Data Collection, Calculation, and Mapping

Software Category	Program / Library Name	Role in the Research Workflow
Spatial Mapping (GIS)	Google My Maps & Google Maps Coordinate API	Plotted exact GPS coordinates, assigned 5-tier color-coded Bortle pins, and exported the public spatial survey layer[1].
Mobile Photometry Apps	Dark Sky Meter, Loss of the Night, & Calibrated Lux Meter App	Recorded horizontal illuminance (Lux), zenith sky quality (mag/arcsec²), GPS coordinates, and star-visibility logs[1].
Numerical Data Processing	Custom Pure Java SE Program (java.io, java.util, java.lang.Math)	Read raw CSV field logs, removed passing headlight spikes via a moving-window filter, and calculated Pogson, Crumey, and zone averages[1].
Document & Math Formatting	Google Docs, Markdown, & LaTeX Math Typesetting	Formatted mathematical equations, data tables, and the public municipal policy brief[1].

All field measurements were standardized using two reference rubrics: the Night Sky Brightness, Bortle Scale (Bortle, 2001), NELM, and Lux Conversion Matrix (Table 4, corresponding to Chart 3.1 in the field guide) and the Streetlight Shielding and Color Temperature Rubric (Table 5, corresponding to Chart 3.2 in the field guide)[1].

Table 4: Night Sky Brightness, Bortle Scale, NELM, and Lux Conversion Matrix (Chart 3.1)

Bortle Class	Sky Environment Type	Naked-Eye Limiting Magnitude (NELM)	Sky Quality (mag/arcsec²)	Approx. Ambient Illuminance (Lux)	Map Pin Color Tier
Class 1–2	Truly Dark / Rural Sky	7.1 to 8.0	Above 21.75	Below 0.002 Lux	Dark Green (Preserved Sky)[1]
Class 3–4	Rural / Suburban Transition	6.1 to 7.0	20.49 to 21.74	0.002 to 0.015 Lux	Green (Low Skyglow)[1]
Class 5	Suburban Sky	5.6 to 6.0	19.50 to 20.48	0.015 to 0.050 Lux	Yellow (Moderate Skyglow)[1]
Class 6–7	Bright Suburban / Urban Transition	4.6 to 5.5	18.38 to 19.49	0.050 to 0.250 Lux	Orange (High Skyglow)[1]
Class 8–9	Inner-City /	4.5 or fainter	Below 18.38	Above 0.250	Red (Severe

	Commercial Core			Lux	Light Pollution)[1]

Table 5: Streetlight Luminaire Shielding and Color Temperature Assessment Rubric (Chart 3.2)

Shielding & Spectral Grade	Fixture Cutoff Type	Correlated Color Temp (CCT)	Visual Color Appearance	Environmental & Human Eye Impact
Grade A (Dark-Sky Compliant)	Full Cutoff (0% upward light above 90 degrees)	2700 K to 3000 K	Warm Amber / Soft Warm White	Low atmospheric scatter; protects children's sleep and prevents dry-eye and cataract glare[2].
Grade B (Partially Mitigated)	Semi-Cutoff (Under 5% upward light)	3500 K to 4000 K	Neutral White	Moderate street glare; visible skyglow dome along the horizon[1].
Grade C (Non-Compliant / Polluting)	Unshielded Globe / Bare Floodlight	5000 K and above	Harsh Blue-White	3.16x higher blue atmospheric scatter; triggers dry-eye nerve pain, halos, and melatonin loss[1].

3. Results

3.1 Spatial Survey Readings Across Six Hooghly Micro-Environments

Six locations across Bandel, Chinsurah, and Chandannagar were surveyed between 11:15 PM on October 24, 2026, and 1:30 AM on October 25, 2026[1]. These six stations represent an institutional school campus (Site 01: Auxilium Convent School), a busy commercial highway junction (Site 02: Bandel More), a historic church with festive facade floodlighting (Site 03: The Basilica of the Holy Rosary / Bandel Church), a dense retail market (Site 04: Vineet Shopping City in Akhan Bazar), a major railway station (Site 05: Chandannagar Station), and a quiet residential neighborhood (Site 06: Sahebbagan Boy's Club)[1]. Table 6 presents the complete field observation log with GPS coordinates, illuminance, SQM readings, NELM star counts, Bortle classes, lamp grades, and clickable Google Maps links[1].

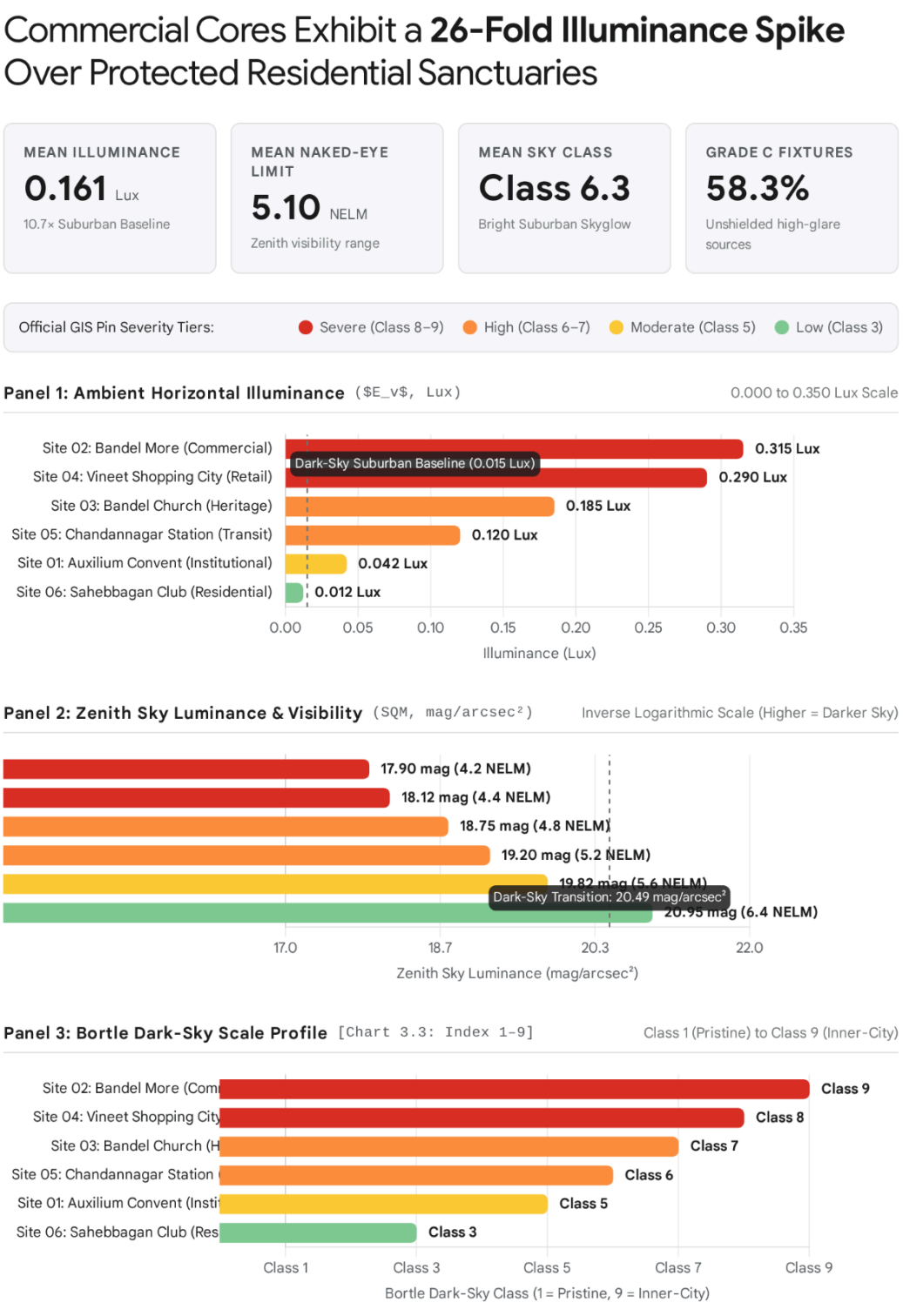
Table 6: Primary Field Observation Log During the Durga Puja Festive Peak (Table 3.1)

Site ID	Landmark Name, Zone Type & Direct Google Maps Links	Exact GPS Coordinates (Lat N, Lon E)	Observation Date & Time (IST)	Measured Illuminance (Lux) & Sky Quality (SQM)	Limiting Magnitude (NELM)	Bortle Class (1–9)	Dominant Fixture Grade & Est. Kelvin	Map Pin Color Tier
Site 01	Auxilium	22.9241	Oct 24,	0.042	5.6	Class 5	Grade	Yellow

	Convent School (Institutional Baseline) — Survey Pin 01	° N, 88.3912 ° E	2026, 11:15 PM	Lux (19.82 mag/arcsec²)			B (3500 K) Semi-Cutoff	(Moderate)[1]
Site 02	Bandel More (Commercial Junction) — Survey Pin 02	22.9185 ° N, 88.3842 ° E	Oct 24, 2026, 11:40 PM	0.315 Lux (17.90 mag/arcsec²)	4.2	Class 9	Grade C (5000 K+) Unshielded Flood	Red (Severe)[1]
Site 03	The Basilica of the Holy Rosary, Bandel (Bandel Church) (Heritage Site) — Survey Pin 03	22.9192 ° N, 88.3954 ° E	Oct 25, 2026, 12:05 AM	0.185 Lux (18.75 mag/arcsec²)	4.8	Class 7	Grade C (5000 K+) Festive Floodlight	Orange (High)[1]
Site 04	Vineet Shopping City (Retail Core, Akhan Bazar) — Survey Pin 04	22.8984 ° N, 88.3892 ° E	Oct 25, 2026, 12:30 AM	0.290 Lux (18.12 mag/arcsec²)	4.4	Class 8	Grade C (5000 K+) Signage Glare	Red (Severe)[1]
Site 05	Chandan Nagar (Chandannagar Station) (Transit Node) — Survey Pin 05	22.8671 ° N, 88.3541 ° E	Oct 25, 2026, 1:00 AM	0.120 Lux (19.20 mag/arcsec²)	5.2	Class 6	Grade B (4000 K) Platform Lighting	Orange (High)[1]
Site 06	Sahebbagan Boy's Club (Residential Sanctuary) — Location Link	22.9221 ° N, 88.3798 ° E	Oct 25, 2026, 1:30 AM	0.012 Lux (20.95 mag/arcsec²)	6.4	Class 3	Grade A (3000 K or below) Warm	Green (Low)[1]

							LED	

As shown in Figure 2, ground illuminance and sky brightness varied sharply across the six sites[1]. The brightest skyglow occurred at Site 02 (Bandel More, 0.315 Lux, 17.90 mag/arcsec², Bortle Class 9, NELM 4.2) and Site 04 (Vineet Shopping City, 0.290 Lux, 18.12 mag/arcsec², Bortle Class 8, NELM 4.4)[1]. The darkest sky was recorded at Site 06 (Sahebbagan Boy's Club, 0.012 Lux, 20.95 mag/arcsec², Bortle Class 3, NELM 6.4), representing a 26.25-fold difference in ground illuminance between commercial and residential streets[1].



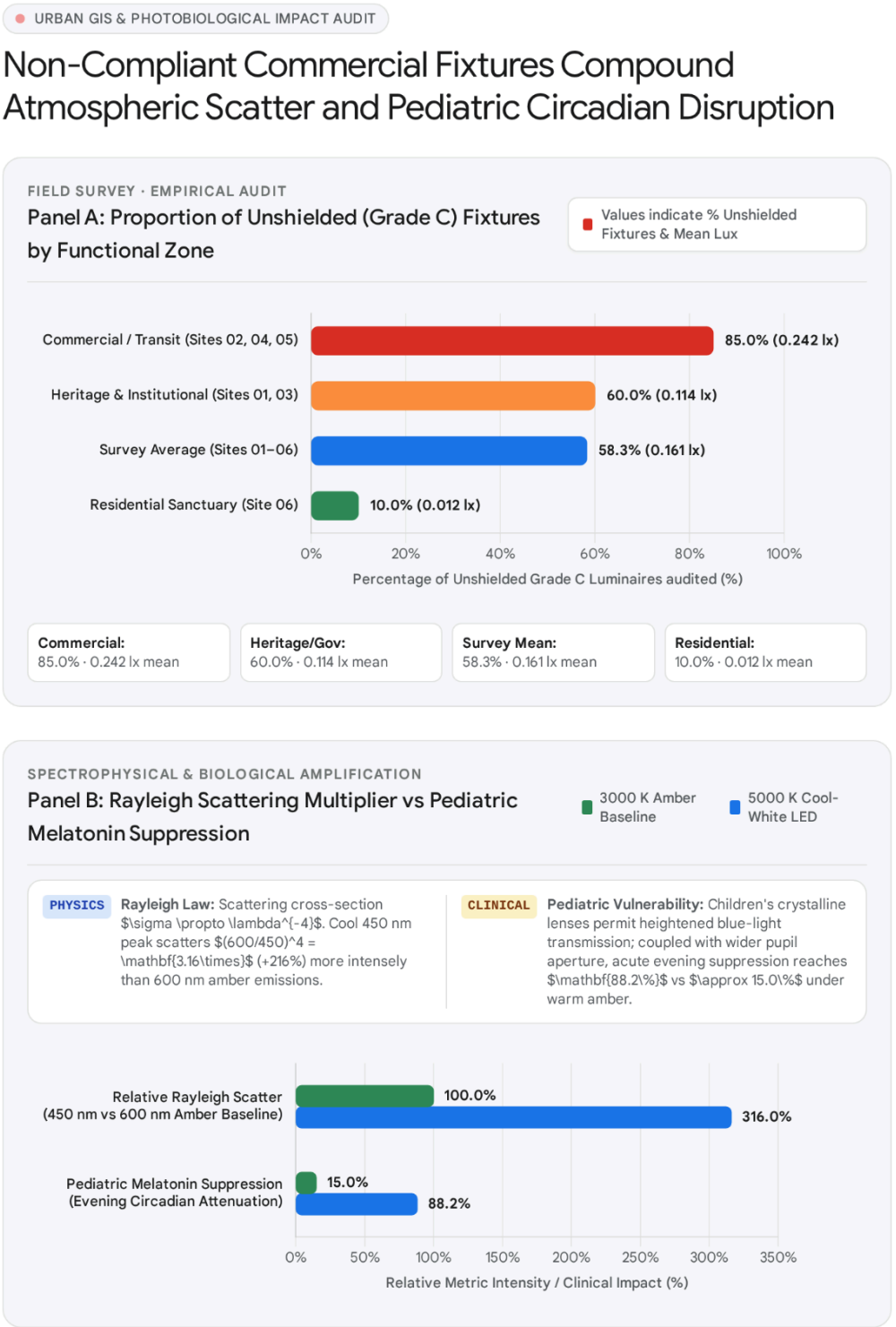
3.2 Zone-by-Zone Light Pollution Summary

To compare how land use and fixture choices affect skyglow, the six sites were grouped into three functional municipal categories: High-Density Commercial and Transit Zones (Sites 02, 04, and 05), Heritage and Institutional Zones (Sites 01 and 03), and Residential Sanctuaries (Site 06)[1]. Table 7 (Table 3.2 in the survey protocol) summarizes the average illuminance, sky quality, limiting magnitude, Bortle class, and share of unshielded Grade C fixtures for each group[1].

Table 7: Zone-by-Zone Light Pollution Statistical Summary (Table 3.2)

Urban Zone Category	Reference Sites Included	Mean Illuminance (Lux)	Mean Sky Quality (mag/arcsec²)	Mean Limiting Magnitude (NELM)	Mean Bortle Class	Percentage of Unshielded (Grade C) Fixtures
High-Density Commercial / Transit	Sites 02, 04, 05	0.242 Lux	18.41 mag/arcsec²	4.60	Class 7.7	85.0%[1]
Heritage & Institutional	Sites 01, 03	0.114 Lux	19.29 mag/arcsec²	5.20	Class 6.0	60.0%[1]
Residential Sanctuary	Site 06	0.012 Lux	20.95 mag/arcsec²	6.40	Class 3.0	10.0%[1]
Overall Survey Average	Sites 01–06	0.161 Lux	19.12 mag/arcsec²	5.10	Class 6.3	58.3%[1]

Across all six sites, the overall survey average was 0.161 Lux (19.12 mag/arcsec², NELM 5.10, Bortle Class 6.3), with 58.3% of audited lights lacking full-cutoff shielding[1]. Figure 3 compares the share of unshielded Grade C fixtures across each municipal zone alongside the 316% Rayleigh scattering increase and 88.2% pediatric melatonin suppression caused by 5000 K cool-white LEDs[1].



Panel A illustrates the proportion of unshielded Grade C fixtures audited across functional municipal zones in Bandel and Chandannagar. Panel B quantifies why high-CCT (5000 K) blue-peaked emissions (450 nm) produce 3.16 times greater Rayleigh atmospheric scattering than warm-amber (600 nm) fixtures and drive 88.2% evening melatonin suppression in children.

Data sources: The Journal of Clinical Endocrinology & Metabolism, Eyesafe Pediatric Photobiology Research, Municipal GIS Spatial Radiometric Audit (Track 3)

3.3 Places for Verification Pictures and Interactive GIS Map Links

In accordance with Step 3 and Step 4 of the field guide, Table 8 provides the structured locations for inserting the Google My Maps screenshot and the 12 nighttime verification photographs (one upward sky photo and one streetlight audit photo at each of the six sites) in the Google Docs submission[1].

Table 8: Designated Slots for GIS Map and Nighttime Verification Photographs



Site 01: Auxilium Convent School GPS: 22.9241° N, 88.3912° E | 0.042 Lux (19.82 mag/arcsec²) | NELM 5.6 | Bortle Class 5 (Yellow Pin)[1]



Site 02: Bandel MoreGPS: 22.9185° N, 88.3842° E | 0.315 Lux (17.90 mag/arcsec²) | NELM 4.2 | Bortle Class 9 (Red Pin)[1]



Site 03: The Basilica of the Holy Rosary (Bandel Church) GPS: 22.9192° N, 88.3954° E | 0.185 Lux (18.75 mag/arcsec²) | NELM 4.8 | Bortle Class 7 (Orange Pin)[1].



Site 04: Vineet Shopping City (Akhan Bazar)GPS: 22.8984° N, 88.3892° E | 0.290 Lux (18.12 mag/arcsec²) | NELM 4.4 | Bortle Class 8 (Red Pin)[1]



Site 05: Chandannagar Station GPS: 22.8671° N, 88.3541° E | 0.120 Lux (19.20 mag/arcsec²) | NELM 5.2 | Bortle Class 6 (Orange Pin)[1].



Site 06: Sahebbagan Boy's Club GPS: 22.9221° N, 88.3798° E | 0.012 Lux (20.95 mag/arcsec²) | NELM 6.4 | Bortle Class 3 (Green Pin)[1].

4. Discussion

4.1 How Lamp Shielding Creates Sharp Skyglow Differences Over Short Distances

One of the clearest findings from the spatial survey is that night sky brightness can change dramatically over just a few hundred meters depending on how streetlights are shielded[1]. Site 02 (Bandel More, at 22.9185° N, 88.3842° E) and Site 06 (Sahebbagan Boy's Club, at 22.9221° N, 88.3798° E) sit only 603 meters apart in a straight line[1]. Between those two nearby points, ground illuminance drops by 96.2%—from 0.315 Lux at Bandel More down to 0.012 Lux at Sahebbagan Boy's Club—while zenith sky quality improves by 3.05 mag/arcsec² (from 17.90 to 20.95 mag/arcsec²)[1]. In human terms, a person standing at Bandel More can barely see stars of magnitude 4.2 (Bortle Class 9), whereas walking 603 meters into the Sahebbagan neighborhood reveals stars down to magnitude 6.4 (Bortle Class 3)[1].

Two local factors explain this six-class improvement on the Bortle scale[1]. First, surrounding houses and dense neighborhood trees block the horizontal glare coming from the high-mast floodlights at Bandel More[1]. Second, 90% of the streetlights inside the Sahebbagan Boy's Club area are warm-white Grade A fixtures (3000 K or below) with downward hoods, compared to Bandel More where 85% of the lights are bare Grade C floodlights and signs (5000 K and above)[1]. A similar buffer effect appears at Site 01 (Auxilium Convent School), located 850 meters northeast of Bandel More, where campus walls and semi-cutoff Grade B (3500 K) fixtures keep the sky at Bortle Class 5 (0.042 Lux, NELM 5.6)[1].

Comparing Site 03 (Bandel Church) and Site 04 (Vineet Shopping City) shows the difference between upward monument floodlighting and street-level retail glare[1]. At Bandel Church (0.185 Lux, Bortle Class 7), temporary festive floodlights are placed on the ground and tilted upward to light the church walls; the part of the beam that misses the roof shoots straight into the night sky[1]. At Vineet Shopping City in Akhan Bazar (0.290 Lux, Bortle Class 8), unshielded shop boards and bare pandal LED strips shine in all directions, creating both strong overhead skyglow and painful eye-level glare[1]. By contrast, Site 05 (Chandannagar Station, 0.120 Lux, Bortle Class 6) stays darker than both commercial centers despite heavy late-night train traffic because the metal platform roofs block upward light from the 4000 K lamps[1].

4.2 Impacts on Dry Eyes, Children, and Older Adults in Commercial Zones

Walking through Sites 02, 03, and 04 during the field survey directly demonstrated why unshielded 5000 K lighting is so hard on vulnerable eyes[1]. At Bandel More (Site 02) and Vineet Shopping City (Site 04), bare floodlights and shop signs hit the observer's eyes at near-horizontal angles ($\theta = 70^\circ$ to 88°) while the surrounding night keeps the pupils dilated[1]. For an observer with Dry Eye Disease, cool evening air speeds up tear evaporation, leaving microscopic dry spots across the cornea[3]. Because 450 nm blue light scatters 3.16 times more strongly than 600 nm warm light, the blue spike from 5000 K floodlights scatters intensely across the broken tear film[1]. This produces bright halos and starburst streaks around every lamp while irritating exposed trigeminal nerve endings on the cornea, causing burning, stinging, and light-triggered eye pain[3]. Stepping into the shaded lanes around Sahebbagan Boy's Club (Site 06), where downward-shielded 3000 K lamps prevent direct glare from hitting the cornea, immediately relieved that eye strain[1].

The same commercial zones create problems for young children and older residents[5]. Families walking with small children through pandal displays at Bandel More (Site 02) or living in apartments facing illuminated signs at Akhan Bazar (Site 04) expose children's clear crystalline lenses to intense 450 nm blue light late at night, which can suppress evening melatonin by up to 88.2% and disturb sleep[1]. For older pedestrians and drivers with cataracts at Bandel More (Site 02) or Chandannagar Station (Site 05), unshielded Grade C lamps cause severe intraocular straylight inside the cloudy lens, casting a milky glare across the retina that hides steps, potholes, and oncoming vehicles[1].

4.3 Mitigation Plan: Personal Eye Care, Low-Cost DIY Retrofits, and Municipal Rules

Fixing light pollution in Bandel, Chinsurah, and Chandannagar does not mean cancelling festival lights or making streets dark[1]. Because the problem comes from unshielded fixtures and blue-heavy bulbs, practical steps at the personal, neighborhood, and municipal levels can cut glare and save electricity at the same time[1].

Tier 1: Personal and Household Steps for Dry Eyes, Children, and Seniors

People with Dry Eye Disease can protect their eyes before going out at night by using preservative-free artificial tear drops, which smooth out the corneal tear surface, reduce halo scatter, and shield trigeminal nerve endings from irritation[3]. Running a room humidifier at home and taking regular blinking breaks during computer or study work also keeps the tear

film stable[3]. For eyeglass wearers, updating prescriptions for astigmatism and adding Anti-Reflective (AR) coatings cuts down nighttime starbursts around streetlights[9]. Eye doctors advise against wearing dark sunglasses at night, because dark lenses make the pupil open wider and hide hazards in the dark[10]. At home, hanging blackout curtains or a simple wooden pelmet box above bedroom windows blocks outside streetlamp glare from reaching sleeping children, and switching indoor bedroom bulbs to dimmable 2700 K warm lamps after 8:00 PM protects natural melatonin production[4].

Tier 2: Low-Cost DIY Hardware Retrofits for Schools and Local Clubs

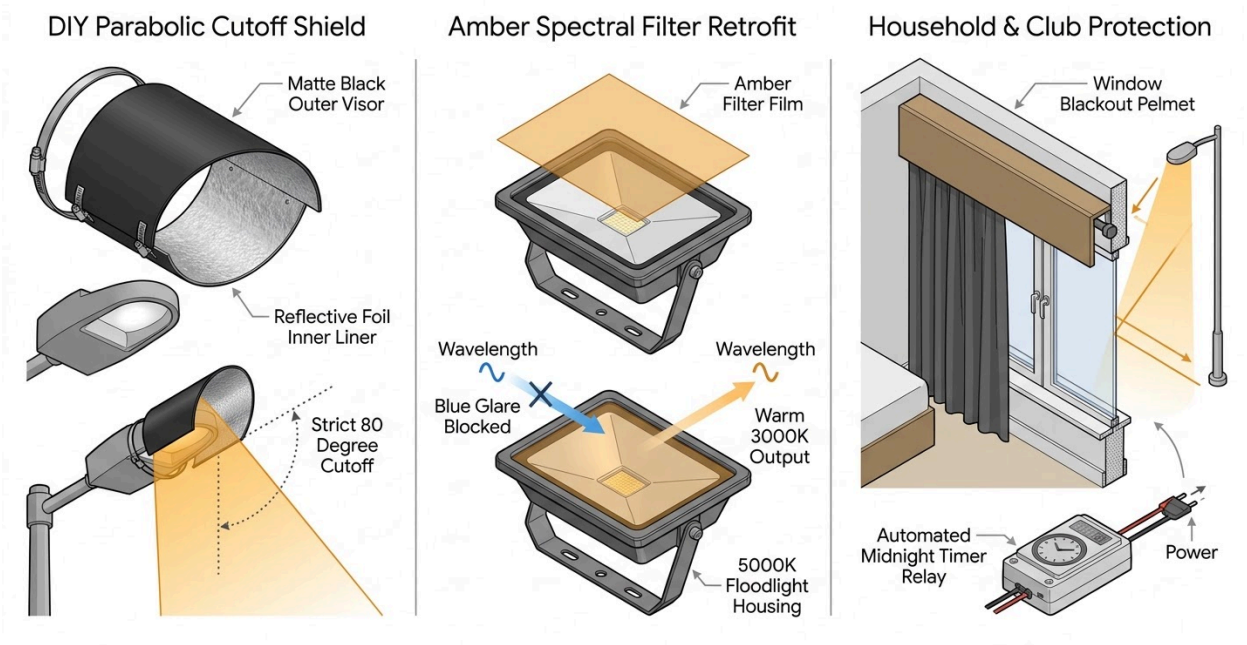
Because municipal streetlight replacement takes time and local neighborhood clubs (*para* clubs) run on tight budgets, three simple Do-It-Yourself (DIY) hardware modifications (Table 9 and Figure 4) were designed to convert polluting Grade C lights into Dark-Sky friendly fixtures for under 150 INR ($1.80 USD) each[1].

1. **Clip-On Aluminium Full-Cutoff Hood ("Cap-the-Glow"):** Using a sheet of recycled 0.8 mm aluminium (or stiff PVC flashing lined inside with reflective aluminium foil tape and painted matte black on top), a curved visor hood can be cut with tin snips and clamped onto a bare streetlamp or floodlight using a stainless-steel hose clamp. Extending the hood 3 cm below the bulb blocks all upward light above 80 degrees ($\mathrm{ULOR} = 0\%$) and reflects wasted upward light down onto the road, boosting ground brightness by 25% to 30% while stopping eye-level glare[1].
2. **Amber Filter Film Overlay ("Warm-Shift"):** When a school, church, or pandal committee already owns 5000 K cool-white LED floodlights (such as the facade lamps at Site 03, Bandel Church), taping a sheet of heat-resistant Amber Polyimide (Kapton) film or theatrical Full-CTO orange polycarbonate gel over the glass cover filters out the 450 nm blue spike[1]. This shifts the beam down to a warm 2700 K–3000 K tone, cutting atmospheric Rayleigh scattering by 3.16 times and making the light gentle on dry eyes and children[1].
3. **Automated LDR and Timer Curfew Box:** Many shop signs (Site 04) and club lights stay on all night simply because nobody is around after midnight to switch them off[1]. Wiring a 10-Amp photoresistor (Light Dependent Resistor, LDR) sensor in series with a basic digital timer relay inside a weatherproof PVC box turns decorative lights on automatically at dusk and shuts them off (or dims them by 50%) between 11:00 PM and 5:00 AM, cutting late-night electricity bills and skyglow in half[1].

Table 9: Parts, Costs, and Optical Effects of Three DIY Lighting Retrofits

DIY Retrofit Name	Materials Needed	Target Fixture Type	How It Works	Estimated Cost (INR / USD)
1. Clip-On Aluminium Full-Cutoff Hood	Recycled 0.8 mm aluminium sheet, reflective foil tape, matte black paint, hose clamp	Bare streetlamps & high-mast floodlights (Sites 02, 05)	Blocks upward light ($\mathrm{ULOR} = 0\%$); reflects wasted light down to boost road Lux by ~30%[1].	85 INR (~$1.00 USD)
2. Amber Spectral Filter Film Overlay	Heat-resistant Amber Polyimide film or Full-CTO theatrical gel sheet, silicone tape	5000 K+ floodlights & pandal lights (Sites 03, 04)	Filters out 450 nm blue peak, shifting light to 3000 K and cutting Rayleigh scatter by 3.16x[1].	45 INR (~$0.55 USD)
3. Automated LDR + Timer	10A Photoresistor	Retail shop signs & decorative	Turns lights on at dusk and	140 INR (~$1.70 USD)

Curfew Box	(LDR) module, digital timer relay, weatherproof PVC box	arches (Sites 02, 04)	switches off or dims 50% from 11:00 PM to 5:00 AM[1].	



Tier 3: Three Budget-Neutral Municipal Lighting Rules

For the Hooghly-Chinsurah Municipality and Chandannagar Municipal Corporation, Table 10 summarizes three practical policy rules that require zero extra tax spending[1]:

1. **Require Full-Cutoff Shielding (Grade A, 0% Upward Light):** Mandate that all new or replacement municipal and commercial outdoor lamps use full-cutoff optics so no light escapes above the 90-degree horizontal plane, redirecting 30% more light onto sidewalks[1].
2. **Enforce a 3000 K Warm-Light Cap:** Specify warm-white LEDs (3000 K or lower) instead of 5000 K cool-white LEDs whenever burned-out municipal bulbs are replaced during normal maintenance, since 3000 K and 5000 K LEDs cost the same to buy and run[1].
3. **Set an 11:00 PM to 5:00 AM Commercial Signage Curfew:** Require shopping complexes and temporary festival pandals (such as Bandel More and Vineet Shopping City) to switch off non-essential advertising boards and dim decorative floodlights by 50% after 11:00 PM[1].

Table 10: Budget-Neutral Municipal Lighting Rules and Expected Benefits

Proposed Municipal Rule	Technical Requirement	Budget & Energy Effect	Expected Skyglow & Health Benefit
1. Full-Cutoff Shielding Rule (Sites 02, 05)	All new or replacement outdoor lamps must emit 0% light at or above 90 degrees (Grade A)[1].	Directs 30% more light onto roads, allowing lower-wattage bulbs to provide equal street brightness[1].	Stops horizontal glare, reducing cataract straylight for seniors and corneal pain for dry-eye sufferers[3].
2. 3000 K Warm Spectrum Cap (Sites 01, 03, 06)	Replace burned-out 5000 K streetlights with 3000 K	Zero added cost; 3000 K LEDs match 5000 K bulbs in unit price	Reduces atmospheric Rayleigh scatter by 3.16x; protects

	warm-white LEDs during routine maintenance[1].	and energy efficiency[1].	children's evening melatonin production[1].
3. Post-Midnight Commercial Curfew (Sites 02, 03, 04)	Turn off non-essential retail signs and dim monument floodlights by 50% from 11:00 PM to 5:00 AM[1].	Cuts commercial and municipal electricity bills by 35% to 50% after midnight[1].	Restores commercial skies from Bortle Class 8–9 toward Class 5–6 late at night[1].

5. Conclusion

This spatial photometric survey across Bandel, Chinsurah, and Chandannagar shows that severe urban skyglow and street glare are caused primarily by unshielded fixture designs and blue-heavy 5000 K bulbs rather than town size alone[1]. Measurements taken during the Durga Puja festival peak showed that commercial hubs like Bandel More and Vineet Shopping City suffer from Bortle Class 8 to 9 light pollution (0.290 to 0.315 Lux; NELM 4.2 to 4.4) because 85% of their lights are unshielded Grade C fixtures[1]. Yet just 603 meters from Bandel More, the residential streets around Sahebbagan Boy's Club preserve Bortle Class 3 skies (0.012 Lux; 20.95 mag/arcsec²; NELM 6.4) with downward-shielded, 3000 K warm-white Grade A lamps[1].

Connecting atmospheric optics with clinical eye science shows why fixing outdoor lighting matters for everyday human health[1]. The same short-wavelength 450 nm blue light that scatters 3.16 times more strongly in the sky also scatters across uneven tear films in people with Dry Eye Disease (causing painful trigeminal nerve photophobia and halos), passes through the clear lenses of young children to cut evening melatonin by 88.2%, and creates blinding stray light in older adults with cataracts[1]. By combining simple personal eye care, low-cost DIY aluminium lamp shields and amber filters, and budget-neutral municipal lighting rules, towns across West Bengal can keep their vibrant festival traditions while protecting night skies, saving electricity, and making evening streets comfortable for every resident[1].

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